

1.FACTORIAL WITH RECURSIVE METHOD:

Recursion is a technique in which a function calls itself repeatedly to solve a problem until it reaches a base condition.

ALGORITHM:

1. Start.
2. Read a number n from the user.
3. Call the factorial function with n .
4. Check if $n \leq 1$.
5. If true, return 1.
6. Otherwise, call the function with $n - 1$.
7. Multiply n with the returned value.
8. Return the result.
9. Display the factorial.
10. STOP.

Time Complexity:

Case	Time Complexity
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Best Case	$O(1)$
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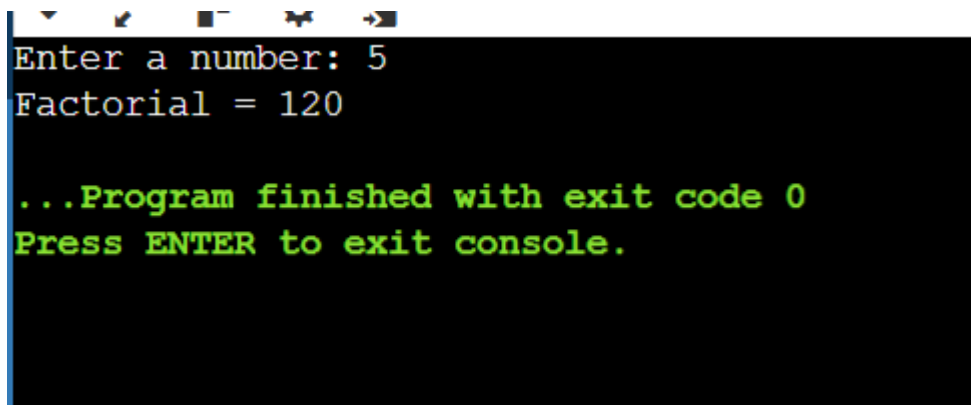
Average Case	$O(n)$
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Worst Case	$O(n)$
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Space Complexity:

Space Complexity = $O(n)$

OUTPUT:

A screenshot of a console window with a black background and white and green text. The text shows the program's execution: it prompts for a number, receives '5', calculates the factorial as 120, and then displays a green message indicating the program finished successfully with exit code 0, followed by a green prompt to press ENTER to exit the console.

```
Enter a number: 5
Factorial = 120

...Program finished with exit code 0
Press ENTER to exit console.
```

Conclusion :

The factorial of a number was successfully calculated using recursion. The function repeatedly calls itself with a smaller value until the base condition is reached. This method is easy to understand for recursive problems, but it uses more memory because of function calls. It has **$O(n)$ time complexity and $O(n)$ space complexity**.